

Topic 2.2 - Change in Linear and Exponential Functions

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Distinguish additive change from proportional change across tables, graphs, formulas, and contexts.
- Construct linear and exponential functions from two values or from stated rates.
- Compare models, interpret intersections, and respect discrete versus continuous domains.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

For each table, decide whether the data are exactly linear, exactly exponential, or neither.

x	0	1	2	3
$A(x)$	14	20	26	32
$B(x)$	14	21	31.5	47.25
$C(x)$	14	19	27	38

Question 2

Core skill

No calculator

Construct the linear function through $(-2, 11)$ and $(5, -10)$.

Question 3

Core skill

No calculator

Construct an exponential function $f(x) = ab^x$ through $(0, 6)$ and $(3, 48)$.

Question 4

Core skill

No calculator

A quantity increases from 250 to 310 in one year. Write both (i) the additive change and (ii) the multiplicative factor and percent change.

Question 5

Core skill

No calculator

The function $P(t) = 900(0.94)^t$ models a quantity after t days. State the initial value, daily percent change, and whether the graph is increasing or decreasing.

Question 6

Equivalent intervals

No calculator

A linear model changes by 18 units every 3 hours. An exponential model changes by a factor of 1.331 every 3 hours. Express each change per hour.

Section B: Application, comparison, and error analysis**Question 7**

Comparison

Calculator permitted

At $x = 0$, a linear model and an exponential model both equal 50. The linear model adds 12 per unit; the exponential model grows by 16% per unit. Build both functions and determine which is larger at $x = 5$.

Question 8

Comparison

Calculator permitted

For $L(x) = 30 + 20x$ and $E(x) = 30(1.25)^x$, find the first whole-number x for which $E(x) > L(x)$.

Question 9

Error analysis

No calculator

A student sees a curve bending upward and concludes that the function is exponential. Give two reasons that the conclusion is not justified.

Question 10

Modeling

No calculator

A water tank contains 680 L and drains at 35 L per minute. Another tank contains 680 L and loses 7% of its contents per minute. Write both models and state a physically reasonable domain for each.

Question 11

Core skill

No calculator

Find the exponential function whose values at $x = 2$ and $x = 5$ are 40 and 135, respectively. Leave the growth factor in exact form.

Question 12

Construction

No calculator

Create one linear and one exponential function that both pass through $(0, 24)$ and $(2, 54)$. Compare their values at $x = 4$.

Section C: AP-style multiple choice**MCQ 1**

No calculator

Which table represents exponential growth?

- A. y : 4, 9, 14, 19
- B. y : 4, 8, 16, 32
- C. y : 4, 7, 12, 19
- D. y : 4, 2, 0, -2

MCQ 2

No calculator

A function increases by 30% whenever x increases by 2. What is the factor for a one-unit increase in x ?

- A. 1.15
- B. 1.30
- C. $\sqrt{1.30}$
- D. 0.70

MCQ 3

No calculator

Which statement correctly compares $L(x) = 200 + 15x$ and $E(x) = 200(0.97)^x$ for $x \geq 0$?

- A. Both increase.
- B. L increases while E decreases.
- C. Both decrease.
- D. E eventually exceeds L .

MCQ 4

No calculator

Two models intersect at $(6, 410)$. Which interpretation is valid?

- A. Both models are correct for all inputs.
- B. At input 6, both models predict output 410.
- C. The real quantity must equal 410.
- D. The functions have the same rate at $x = 6$.

Section D: Free-response reasoning

FRQ 1 - Build and compare two models

Calculator permitted

6 points

A delivery service offers two pricing plans. Plan L charges 90 QAR plus 14 QAR per delivery. Plan E charges 90 QAR and multiplies the total charge by 1.12 for each delivery. (a) Write both models. (b) Compare the predicted charges for 5 deliveries. (c) Determine the first whole-number delivery count for which Plan E exceeds Plan L. (d) Discuss one limitation of each model.

FRQ 2 - Same data, different structures

Calculator permitted

6 points

Two functions pass through $(0, 30)$ and $(4, 150)$. One is linear and one is exponential. (a) Construct both functions. (b) Find each value at $x = 8$. (c) Explain why the functions agree at two inputs but produce very different predictions later. (d) Give a context in which one model would be more defensible than the other.
